Fabrice Lamarche

PhD Candidate, Artificial Intelligence — Mila

fabricelamarche0@gmail.com
linkedin.com/fabrice-lamarche
github.com/fbl10
fabricelamarche.com

Professional experience

Intact Insurance

Data Scientist — 2024 – 2025

- Redefined the evaluation process for market value prediction
- Performance analysis, model maintenance

RemedIA — IVADO

AI Consultant – Scientist in Residence — 2023 – 2025

- EdTech startup: AI specialist
- Fine-tuning and RAG setup for LLM tutoring

Government of Québec, CIUSSS Centre-Sud

AI Analyst — 2021 – 2023

- Developed an XGBoost model for hospital stay duration (< 1 RMSE)
- Analysis and visualization of strategic data

Publications

UMCR: Universal Medical Concept Recognition through Guideline Integration and Agentic Linking

F. Lamarche and A. Zouaq — *arXiv preprint* 2026.

- Universal pipeline for medical concept recognition over any target ontology and any source text type
- Direct guideline integration and agentic linking module
- Reduces dependence on parametric LLM memory

MedHal: a Synthetic Dataset for Medical Hallucination Detection

F. Lamarche et al. — *AACL-IJCNLP 2026* (to appear).

- Synthetic dataset allowing the identification of hallucinations in medical text
- Data creation and filtering, fine-tuning, evaluation and expert validation

BenchIE^{FL}: A Manually Re-Annotated Fact-Based Open Information Extraction Benchmark

Fabrice Lamarche and Philippe Langlais — *ACL 2024 Findings*.

- Manual annotation of OpenIE data, creation of an automated evaluation metric

Education

Mila — Polytechnique Montréal

PhD Candidate — Artificial Intelligence (LAMA-WeST Lab) — 2024 – present

- Language models, nonparametric memories and synthetic data for medical text processing

Université de Montréal, Department of Computer Science and Operations Research (DIRO)

Master's degree — Artificial Intelligence (RALI) — 2021 – 2023

- DIRO Writing Fellowship 2023
- Natural Language Processing (NLP)
- Open information extraction (OpenIE) and evaluation methods

Bachelor's degree — Mathematics and Computer Science

Data science speciality — 2018 – 2021

Presentations

IVADO — Raising awareness on AI

2025 – 2026

- The ABCs of AI
- AI, a risk for our democracy?

LAMA-WeST Seminars

2025

- BenchIE^{FL}, MedHal, UMCR, DRAC coding-assistant

Competitions

Datathon 2022, 2023 — PolyFinance

AI and NLP competition applied to finance.

- Exchange rate prediction (time series)
- Best performing solution using FinBERT + CNN-LSTM (0.0004 MSE)

CodeML 2022 — PolyAI

Machine Learning Competition, Polytechnique Montréal.

Skills

Research

- Evaluation of ML models
- LLM fine-tuning, RAG, in-context learning, agentic workflows
- RL

Python

- PyTorch, Transformers, vLLM, Ollama

Cloud

- Azure ML, Snowflake, SQL

Languages

English, French